

Number Systems -- Guided Notes

Unit 1.3 | CompTIA Network+ N10-009 Obj. Core 1 (foundational) | N10-008

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Binary is a base-_____ number system using only the digits _____ and _____. Each position represents a power of two. Fill in the missing position values from left to right:

128 | ____ | ____ | 16 | ____ | 4 | ____ | 1

Question 2

Convert the decimal number 157 to binary. Use the subtraction method: start at the highest position (128), subtract if it fits, and record a 1; otherwise record a 0. Show your work step by step.

Question 3

Hexadecimal is a base-_____ number system. It uses the digits 0-9 and the letters _____. The letter A = _____, B = _____, C = _____, D = _____, E = _____, F = _____.

Question 4

To convert binary to hexadecimal, group the binary digits into sets of _____. For example, convert 11001010 to hex:

1100 = _____ | 1010 = _____

Result in hex: _____

Question 5

The hexadecimal value 0xFF equals _____ in decimal. In binary, this is _____ (all eight bits set to 1). This is the maximum value that can be stored in _____ byte of memory.

Question 6

Data transfer speeds are measured in _____ (bits per second). Storage sizes are measured in _____ (bytes). Since 1 byte = _____ bits, a 1 Gbps (gigabit per second) connection transfers data at _____ MB/s (megabytes per second).

Question 7

The prefix _____ (e.g., 0xFF) indicates a hexadecimal value. The APIPA address range _____ . _____ .x.x is assigned to a device when it cannot reach a DHCP server. A MAC address is a 48-bit address written as _____ hexadecimal pairs (e.g., AA:BB:CC:DD:EE:FF).

Question 8 -- Real World Application

REAL WORLD: A network technician reads a MAC address filter log and sees the entry: 00:1A:2B:3C:4D:5E. Convert the last octet (5E) from hexadecimal to binary and then to decimal. Explain why hex is used for MAC addresses instead of binary or decimal.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/1-3-number-systems/1-3-number-systems-answer-key.pdf
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